

User Acceptance Testing (UAT) Template

Date	30 September 2026
Team ID	SWTID-2026-9682
Project Name	AI StudyBuddy – AI-Augmented Backend Development
Maximum Marks	

Project Overview:

Project Name: AI StudyBuddy – AI-Augmented Backend Development

Project Description: AI-powered learning platform backend (Node.js, Express.js, MongoDB, JWT, Google Gemini API) that lets students upload study material and generate summaries, flashcards, quizzes and personalised study plans.

Project Version: v1.0

Testing Period: 26.09.2026 to 27.09.2026

Testing Scope:

Features to be tested:

- User registration and login (JWT authentication, bcrypt password hashing)
- Role-based access control (Student / Admin)
- Study material upload and management
- AI summary, flashcard, quiz and study plan generation (Google Gemini API)
- Storage and retrieval of generated resources (MongoDB)
- Admin user management and usage monitoring

User stories / requirements to be tested:

- As a student, I can register and log in securely.
- As a student, I can upload study material and get an AI summary.
- As a student, I can generate flashcards and an MCQ quiz from my material.
- As a student, I can get a study plan based on my exam date.
- As a student, I can view my previously generated resources.
- As an admin, I can manage users and monitor system usage.

Testing Environment:

URL/Location: mongodb://localhost:27017/

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Register with valid details	1. Open Register page / POST /register 2. Enter name, unique email, password 3. Submit	Account created; password stored hashed; success response returned.	To be recorded after testing	Pending
TC-002	Register with an existing email (negative)	1. Register with an email already in the database 2. Submit	Request rejected with a duplicate-email error; no new user created.	To be recorded after testing	Pending
TC-003	Login with valid credentials	1. Open Login page / POST /login 2. Enter registered email and password 3. Submit	JWT token issued and user redirected to dashboard.	To be recorded after testing	Pending
TC-004	Login with wrong password (negative)	1. Enter valid email with incorrect password 2. Submit	Login denied with an invalid-credentials error; no token issued.	To be recorded after testing	Pending
TC-005	Access protected route without token (negative)	1. Call a protected API (e.g., generate summary) with no Authorization header	401 Unauthorized returned; AI service is not called.	To be recorded after testing	Pending
TC-006	Upload study material	1. Log in as student 2. Enter title, subject and content 3. Submit	Material saved in MongoDB and shown in the	To be recorded after testing	Pending

			student's material list.		
TC-007	Generate AI summary	1. Select an uploaded material 2. Click Generate Summary	Concise summary generated by Gemini, displayed and stored against the material.	To be recorded after testing	Pending
TC-008	Generate flashcards	1. Select a material 2. Click Generate Flashcards	Multiple question–answer flashcards generated and saved.	To be recorded after testing	Pending
TC-009	Generate quiz	1. Select a material 2. Click Generate Quiz	MCQ quiz with options and answers generated and saved.	To be recorded after testing	Pending
TC-010	Generate study plan	1. Enter subject, exam date and available study time 2. Click Generate Study Plan	Personalised day-wise plan generated up to the exam date and saved.	To be recorded after testing	Pending
TC-011	AI request with empty content (negative)	1. Submit an AI generation request with empty content	Validation error returned; no Gemini call made and nothing saved.	To be recorded after testing	Pending
TC-012	View saved resources	1. Log in 2. Open dashboard / history	Previously generated summaries, flashcards, quizzes and plans are	To be recorded after testing	Pending

			listed for that user only.		
TC-013	Student accesses admin route (negative, RBAC)	1. Log in as student 2. Call an admin API (e.g., list all users)	403 Forbidden returned.	To be recorded after testing	Pending
TC-014	Admin manages users	1. Log in as admin 2. View user list 3. Update or remove a user	Admin can view and manage users; changes reflected in the database.	To be recorded after testing	Pending
TC-015	Password stored securely	1. Register a user 2. Inspect the user document in MongoDB	Password field contains a bcrypt hash, never plain text.	To be recorded after testing	Pending

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	No bugs logged yet – record issues found during testing here	—	—	—	—

Sign-off:

Tester Name: Vignesh.B

Date: 27.09.2026

Notes:

- Ensure that all test cases cover both positive and negative scenarios.

- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.

References:

Postman Learning Center (API testing): <https://learning.postman.com/docs/>

JSON Web Token Introduction: <https://jwt.io/introduction>

Google Gemini API Documentation: <https://ai.google.dev/gemini-api/docs>

Express.js Routing Guide: <https://expressjs.com/en/guide/routing.html>

MongoDB Documentation: <https://www.mongodb.com/docs/>